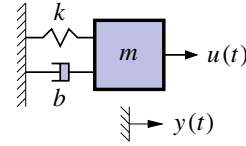




An example to use with nonlinear KFs

- To illustrate nonlinear Kalman filters, we will modify the spring-mass-damper example we have used to this point.
- The first change is to replace the linear spring with a nonlinear spring.
- If we let $y(t)$ be deviation in position away from equilibrium, then a linear spring has $F(t) = ky(t)$; a nonlinear spring has $F(t) = k_1 y(t) + k_2 y^3(t)$ and:

$$m\ddot{y}(t) = u(t) - b\dot{y}(t) - k_1 y(t) - k_2 y^3(t).$$



- We cannot put this in a linear state-space form. The closest that we come is:

$$\begin{bmatrix} \dot{y}(t) \\ \ddot{y}(t) \end{bmatrix} = \begin{bmatrix} \dot{y}(t) \\ -\frac{b}{m}\dot{y}(t) - \frac{k_1}{m}y(t) - \frac{k_2}{m}y^3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} (u(t) + w(t)),$$

for one way of adding process noise $w(t)$, if $x_1(t) = y(t)$ and $x_2(t) = \dot{y}(t)$.



Converting state equation to discrete time?

- According to this representation, $x(t) = \begin{bmatrix} y(t) & \dot{y}(t) \end{bmatrix}^T$ and we have $\dot{x}(t) = f_c(x(t), u(t), w(t))$ if we let:

$$f_c(x(t), u(t), w(t)) = \begin{bmatrix} \dot{y}(t) \\ -\frac{b}{m}\dot{y}(t) - \frac{k_1}{m}y(t) - \frac{k_2}{m}y^3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} (u(t) + w(t)).$$

- We cannot use the same approach as we used for linear models to convert the continuous-time state-equation function $f_c(\cdot)$ to discrete time.
- Instead, I will approximate the derivatives using Euler's rule:

$$\begin{aligned} \begin{bmatrix} \frac{y(t+\Delta t) - y(t)}{\Delta t} \\ \frac{\dot{y}(t+\Delta t) - \dot{y}(t)}{\Delta t} \end{bmatrix} &\approx \begin{bmatrix} \dot{y}(t) \\ -\frac{b}{m}\dot{y}(t) - \frac{k_1}{m}y(t) - \frac{k_2}{m}y^3(t) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{m} \end{bmatrix} (u(t) + w(t)) \\ \begin{bmatrix} y_{k+1} \\ \dot{y}_{k+1} \end{bmatrix} &\approx \begin{bmatrix} y_k + \Delta t (\dot{y}_k) \\ \dot{y}_k - \Delta t \left(\frac{b}{m}\dot{y}_k + \frac{k_1}{m}y_k + \frac{k_2}{m}y_k^3 \right) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\Delta t}{m} \end{bmatrix} (u_k + w_k). \end{aligned}$$



Computing $\hat{A}_k$

- So, in discrete time, our approximate model has

$x_{k+1} = f(x_k, u_k, w_k)$, where:

$$f(x_k, u_k, w_k) = \begin{bmatrix} x_{1,k} + \Delta t (x_{2,k}) \\ x_{2,k} - \Delta t \left(\frac{b}{m}x_{2,k} + \frac{k_1}{m}x_{1,k} + \frac{k_2}{m}x_{1,k}^3 \right) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\Delta t}{m} \end{bmatrix} (u_k + w_k).$$

- To implement the EKF, we must find $\hat{A}_k$. Recall that:

$$\hat{A}_k = \left. \frac{df(x_k, u_k, w_k)}{dx_k} \right|_{x_k = \hat{x}_k^+} = \left. \frac{dx_{k+1}}{dx_k} \right|_{x_k = \hat{x}_k^+}.$$

- With the given definition of x_k ,

$$\frac{dx_{k+1}}{dx_k} = \begin{bmatrix} \frac{dx_{1,k+1}}{dx_{1,k}}, & \frac{dx_{1,k+1}}{dx_{2,k}} \\ \frac{dx_{2,k+1}}{dx_{1,k}}, & \frac{dx_{2,k+1}}{dx_{2,k}} \end{bmatrix} \quad \text{and} \quad \hat{A}_k = \begin{bmatrix} 1, & \Delta t \\ -\frac{k_1 \Delta t}{m} - \frac{3k_2(\Delta t)(\hat{x}_{1,k}^+)^2}{m}, & 1 - \frac{b \Delta t}{m} \end{bmatrix}.$$



Computing $\hat{B}_k$

- For convenience, I repeat our discrete-time approximate model

$x_{k+1} = f(x_k, u_k, w_k)$, where:

$$f(x_k, u_k, w_k) = \begin{bmatrix} x_{1,k} + \Delta t (x_{2,k}) \\ x_{2,k} - \Delta t \left(\frac{b}{m} x_{2,k} + \frac{k_1}{m} x_{1,k} + \frac{k_2}{m} x_{1,k}^3 \right) \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{\Delta t}{m} \end{bmatrix} (u_k + w_k).$$

- To implement the EKF, we must also find $\hat{B}_k$. Recall that:

$$\hat{B}_k = \left. \frac{df(x_k, u_k, w_k)}{dw_k} \right|_{w_k = \bar{w}_k} = \left. \frac{dx_{k+1}}{dw_k} \right|_{w_k = \bar{w}_k}.$$

- With the given definition of x_k and zero-mean w_k :

$$\frac{dx_{k+1}}{dw_k} = \begin{bmatrix} \frac{dx_{1,k+1}}{dw_k} \\ \frac{dx_{2,k+1}}{dw_k} \end{bmatrix} \quad \text{and} \quad \hat{B}_k = \begin{bmatrix} 0 \\ \frac{\Delta t}{m} \end{bmatrix}.$$



A nonlinear measurement equation

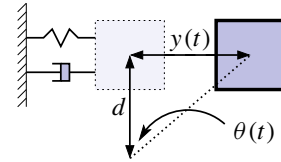
- We also choose to incorporate a nonlinear measurement equation in the model used to illustrate nonlinear KFs.
- Suppose that we measure the angle between the equilibrium block position and its present position from a distance d away from the center of mass:

$$z(t) = \theta(t) + v(t) = \text{atan} \left(\frac{y(t)}{d} \right) + v(t),$$

if we assume that measurement noise is additive.

- A discrete-time model is:

$$z_k = h(x_k, u_k, v_k) = \text{atan} \left(\frac{x_{1,k}}{d} \right) + v_k.$$



Computing $\hat{C}_k$ and $\hat{D}_k$

- To implement the EKF, we must still find $\hat{C}_k$ and $\hat{D}_k$.
- First, we recall that:

$$\hat{C}_k = \left. \frac{dh(x_k, u_k, v_k)}{dx_k} \right|_{x_k = \hat{x}_k^-} = \begin{bmatrix} \frac{d\theta_k}{dx_{1,k}} & \frac{d\theta_k}{dx_{2,k}} \end{bmatrix} \bigg|_{x_k = \hat{x}_k^-} = \begin{bmatrix} \frac{d}{(d)^2 + (\hat{x}_{1,k}^-)^2}, & 0 \end{bmatrix}.$$

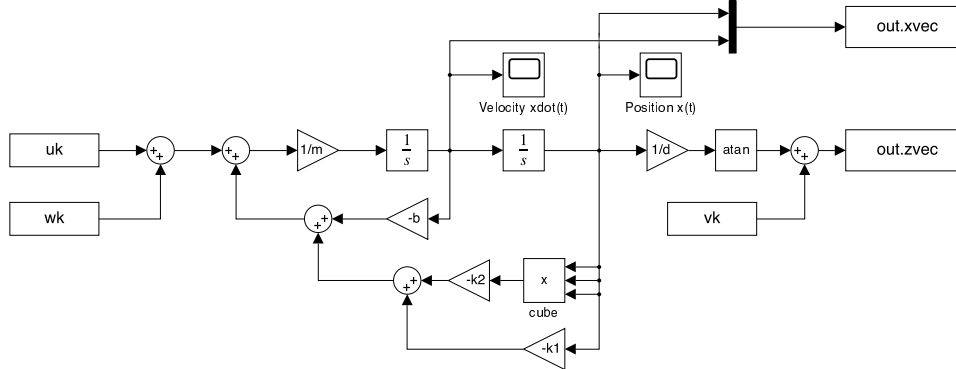
- Finally, we recall that:

$$\hat{D}_k = \left. \frac{dh(x_k, u_k, v_k)}{dv_k} \right|_{v_k = \bar{v}_k} = \left. \frac{d\theta_k}{dv_k} \right|_{v_k = \bar{v}_k} = 1.$$



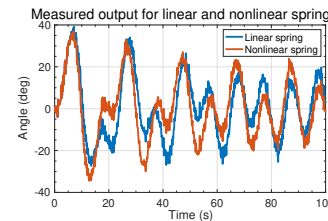
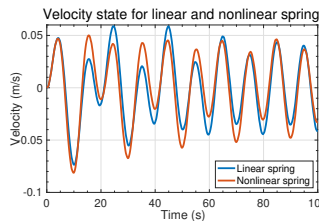
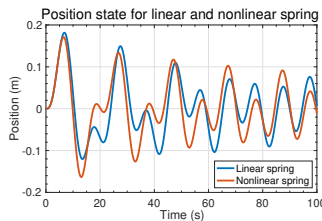
Simulating the nonlinear model

- To create a dataset, we could simulate the discrete-time approximate model using a “for” loop in Octave.
- Instead, I chose to simulate the actual continuous-time model in Simulink:



Sample output

- For the purpose of example, let $m = 50$, $b = 2$, and $k_1 = 4$ (same as before) and $k_2 = 60$, $d = 0.25$.
- The figures below show sample output comparing the linear and nonlinear models for the same input signals.
- We see clear differences between the linear and nonlinear model results.
 - It should be interesting to see nonlinear KF state estimates!



Noise covariances: $\Sigma_{\tilde{v}}$

- One final concern: We need to determine $\Sigma_{\tilde{w}}$ and $\Sigma_{\tilde{v}}$ before we can implement an EKF for this system.
- For sensor noise, we use the result from Lesson 1.3.7 and write $\Sigma_{\tilde{v}} \approx \frac{S_v}{\Delta t}$.
- For process noise, Lesson 1.3.7 would have us either compute

$$\Sigma_{\tilde{w}} \approx (B_w S_w B_w^T) \Delta t = \begin{bmatrix} 0 & 0 \\ 0 & S_w \frac{\Delta t}{m^2} \end{bmatrix},$$

or use the “little trick” involving a matrix exponential to compute $\Sigma_{\tilde{w}}$.

- Either way, the result is 2×2 where EKF Step 1b expects a scalar result.
- Otherwise, the computation $\hat{B}_k \Sigma_{\tilde{w}} \hat{B}_k^T$ does not make sense dimensionally.
- The conflict arises because the state equation used in Lesson 1.3.7 was $x_{k+1} = Ax_k + Bu_k + w_k$ in which w_k has the same dimension as x_k .
 - In our nonlinear example, $w(t)$ and w_k are assumed to be scalars.



Noise covariances: $\Sigma_{\tilde{w}}$

- Note that the computation $\hat{B}_k \Sigma_{\tilde{w}} \hat{B}_k^T$ in EKF Step 1b for this example (assuming scalar $\Sigma_{\tilde{w}}$) results in:

$$\hat{B}_k \Sigma_{\tilde{w}} \hat{B}_k^T = \begin{bmatrix} 0 & 0 \\ 0 & \Sigma_{\tilde{w}} \left(\frac{\Delta t}{m}\right)^2 \end{bmatrix} \quad \text{since} \quad \hat{B}_k = \begin{bmatrix} 0 \\ \frac{\Delta t}{m} \end{bmatrix}.$$

- Compare this result to the 2×2 matrix approximate result for $\Sigma_{\tilde{w}}$ from Lesson 1.3.7,

$$\Sigma_{\tilde{w}} \approx (B_w S_w B_w^T) \Delta t = \begin{bmatrix} 0 & 0 \\ 0 & S_w \frac{\Delta t}{m^2} \end{bmatrix}.$$

- They are identical if we assign (scalar) $\Sigma_{\tilde{w}} = S_w / \Delta t$. We learn that $\hat{B}_k \Sigma_{\tilde{w}} \hat{B}_k^T$ in EKF Step 1b is the total effect of process noise on the prediction-error covariance.
- So, in our EKF, we can: (1) Set $\Sigma_{\tilde{w}} = S_w / \Delta t$ and use the EKF equations unmodified; or (2) Replace $\hat{B}_k \Sigma_{\tilde{w}} \hat{B}_k^T$ with either the 2×2 matrix computed by the “little trick” or its approximate result on this slide.



Summary

- This lesson has defined a nonlinear dynamic system that we will use to illustrate the operation of the EKF.
- To retain some connection with the example we have used already to demonstrate linear KFs, we slightly modify the spring-mass-damper system:
 - We replace the linear spring with a nonlinear spring;
 - We use an angle sensor to measure a function of mass position.
- You learned how to develop the continuous-time equations of motion and how to approximate them in discrete time.
- You then saw how we can compute $\hat{A}_k$, $\hat{B}_k$, $\hat{C}_k$, and $\hat{D}_k$ for this model.
- You saw how to simulate a dataset and observed that the nonlinear model produces signals that are noticeably different from those of the linear model.
- We're now ready to perform state estimation on this dataset using the EKF!